

B.TECH 1ST YEAR • 1ST SEMESTER • ENGINEERING CHEMISTRY

Unit IV

Modern Engineering Materials

COMPLETE FORMULA & REVISION SHEET

⚙ Composites

🔥 Refractories

📖 Lubricants

🏗 Portland Cement

📊 Properties

💡 Flashcards

🎯 Exam Tips

LAST-DAY REVISION WEAPON • EXAM-READY • ALL KEY CONCEPTS



COMPOSITES

DEFINITION: A composite is an engineered material made from two or more constituent materials with significantly different physical or chemical properties that, when combined, produce a material with characteristics different from individual components.

Key formula: Composite = **Matrix** + **Reinforcement**

MATRIX

- Continuous phase (binder)
- Surrounds & holds reinforcement
- Transfers load to reinforcement
- Types: Polymer, Metal, Ceramic

REINFORCEMENT

- Discontinuous/dispersed phase
- Provides strength & stiffness
- Types: Particles, Fibres, Flakes
- Must bond well with matrix

CLASSIFICATION OF COMPOSITES

TYPE	SUB-TYPES	REINFORCEMENT	EXAMPLE	APPLICATION
Particle-Reinforced	Large-particle; Dispersion-strengthened	Particles (coarse/fine)	Concrete, WC-Co (Cermets)	Road paving, cutting tools
Fibre-Reinforced	Continuous (aligned); Discontinuous (chopped)	Glass, Carbon, Kevlar fibres	GFRP, CFRP	Aerospace, sports goods
Structural	Laminates; Sandwich panels	Layers of materials	Plywood, Honeycomb panels	Aircraft floors, walls

KEY PROPERTIES

Mechanical

- High strength-to-weight ratio
- High stiffness
- Fatigue resistance

Chemical

- Excellent corrosion resistance
- Chemical inertness
- Weather resistance

Thermal

- Thermal stability
- Low thermal expansion
- Good insulation

IMPORTANT COMPOSITE TYPES – DETAIL

GFRP – Glass Fibre Reinforced Polymer

Matrix: Polyester/Epoxy | Reinf: Glass fibres

- ✓ Lightweight, cheap, corrosion resistant
- ✓ Used in: boats, car bodies, pipes

MMC – Metal Matrix Composite

Matrix: Al/Ti | Reinf: SiC/Al₂O₃

- ✓ High temp stability, wear resistance
- ✓ Used in: aerospace, automotive parts

CFRP – Carbon Fibre Reinforced Polymer

Matrix: Epoxy | Reinf: Carbon fibres

- ✓ Extremely high strength & stiffness
- ✓ Used in: aircraft, racing cars, prosthetics

CMC – Ceramic Matrix Composite

Matrix: SiC/Al₂O₃ | Reinf: Ceramic fibres

- ✓ High hardness, temperature resistance
- ✓ Used in: turbine blades, brakes

APPLICATIONS SUMMARY

INDUSTRY	COMPOSITE USED	WHY?
Aerospace	CFRP, MMC	Lightweight + high strength

INDUSTRY	COMPOSITE USED	WHY?
Automotive	GFRP, CFRP	Weight reduction, fuel efficiency
Construction	Concrete (particle composite)	Compressive strength
Sports	GFRP, CFRP	Light + stiff equipment
Marine	GFRP	Corrosion & water resistance

REFRACTORIES

DEFINITION: Refractories are materials that can withstand very high temperatures (above 1500°C) without melting, softening, or reacting chemically, and are used to line furnaces, kilns, and reactors.

Key property: **Melting Point > 1500°C** / Must be chemically inert at operating temperature

CLASSIFICATION OF REFRACTORIES

TYPE	COMPOSITION	EXAMPLES	NATURE	USED WITH
Acidic	Silica (SiO ₂), Zirconia (ZrO ₂)	Silica brick, Ganister	Resist acid slags; attacked by basic slags	Glass furnaces, acid linings
Basic	MgO, CaO, Dolomite	Magnesite brick, Chrome-magnesite	Resist basic slags; attacked by acid slags	Steel furnaces, cement kilns
Neutral	Al ₂ O ₃ , Cr ₂ O ₃ , Carbon	Fire clay, Chromite brick, Graphite	Resist both acid and basic slags	Between acid & basic zones

PROPERTIES OF REFRACTORIES

PROPERTY	DEFINITION	SIGNIFICANCE
Refractoriness	Ability to withstand high temperature without fusing/softening	Primary requirement
Melting Point	Temperature at which material begins to melt	Must be > operating temp
Thermal Shock Resistance	Ability to withstand rapid temperature changes without cracking	Prevents cracking in cyclic ops
Porosity	% of void space in the refractory body	Low porosity = higher strength & slag resistance
Mechanical Strength	Resistance to deformation under load at high temp	Must support weight of structure
Chemical Inertness	Resistance to attack by molten metals, slag, gases	Prevents contamination of melt
Thermal Conductivity	Heat transfer through the material	Low = better insulation
Spalling Resistance	Resistance to surface peeling due to temp gradient	Prevents premature failure

FACTORS AFFECTING REFRACTORY LIFE

- ▶ **Temperature:** Higher temp accelerates wear & fusion
- ▶ **Mechanical Load:** Heavy loads cause deformation at high temp
- ▶ **Slag Action:** Molten slag chemically attacks lining
- ▶ **Thermal Cycling:** Repeated heating-cooling causes spalling
- ▶ **Chemical Environment:** Acidic/basic gases cause chemical erosion
- ▶ **Abrasion:** Solid particles erode the surface
- ▶ **Porosity:** High porosity → slag penetration → failure

MANUFACTURING – STEPS

1. Selection of raw materials
- ▶ 2. Crushing & grinding
- ▶ 3. Mixing with binders (clay, water)
- ▶ 4. Moulding/Pressing to shape
- ▶ 5. Drying (remove moisture)
- ▶ 6. Firing at high temperature
- ▶ 7. Testing & quality check

APPLICATIONS

IRON & STEEL FURNACES

CEMENT KILNS

GLASS FURNACES

NON-FERROUS SMELTING

COMPARISON: ACIDIC VS BASIC VS NEUTRAL

FEATURE	ACIDIC	BASIC	NEUTRAL
Chief Constituent	SiO ₂ (>93%)	MgO (>85%)	Al ₂ O ₃ , Cr ₂ O ₃
Example	Silica brick	Magnesite brick	Fire clay brick
Reacts with	Basic slags	Acidic slags	Neither
Melting Point	~1710°C	~2800°C (MgO)	~1840°C
Thermal Shock	Poor	Moderate	Good

LUBRICANTS

DEFINITION: A lubricant is a substance introduced between two moving surfaces to reduce friction, minimize wear and tear, and remove heat generated due to friction.

Functions: Reduce friction | Minimize wear | Heat removal | Prevent corrosion | Acts as sealant | Carry away contaminants

CLASSIFICATION OF LUBRICANTS

TYPE	SUB-TYPES	EXAMPLES	APPLICATIONS
Liquid (Oils)	Animal, Vegetable, Mineral (petroleum), Synthetic	Engine oil, Olive oil, SAE oils, Silicon oil	IC engines, turbines, hydraulic systems
Solid	Layered lattice structure solids	Graphite, Molybdenum disulfide (MoS ₂), Talc, Mica	High temp/vacuum/extreme pressure operations
Semi-solid (Greases)	Soap-thickened oils	Calcium grease, Lithium grease, Vaseline	Slow-speed/heavy load bearings, open gears
Gaseous	Air, Gas bearings	Air, CO ₂ , Steam	High-speed precision equipment, dentist drills

MECHANISMS OF LUBRICATION

MECHANISM	FILM THICKNESS	CONDITION	HOW IT WORKS	EXAMPLE
Hydrodynamic (Fluid film)	Thick film (25–2500 μm)	High speed, low load	Lubricant forms complete film; no metal contact	Journal bearings, crankshafts
Boundary	Very thin (monomolecular)	Low speed, high load	Adsorbed layer of lubricant on metal surface	Gear teeth, slow-moving parts
Extreme Pressure (EP)	Solid reaction film	Very high load/temp	EP additives (S, Cl, P) react with metal forming protective layer	Hypoid gears, heavy machinery
Mixed Film	Intermediate	Transition regime	Combination of hydrodynamic & boundary	Starting/stopping of engines

PROPERTIES OF LUBRICATING OILS

PROPERTY	DEFINITION	SIGNIFICANCE / UNIT
Viscosity	Resistance of a fluid to flow (internal friction)	Critical property; measured in cP or cSt; too high or low = poor lubrication
Viscosity Index (VI)	Rate of change of viscosity with temperature	Higher VI = less change with temp = better oil; Range: 0–100+
Flash Point	Lowest temp at which vapours ignite momentarily when flame is applied	Safety parameter; higher FP = safer at high temp; must be > operating temp
Fire Point	Temp at which vapours burn continuously for ≥5 sec	Always higher than flash point; Fire point > Flash point (by ~5–10°C)
Cloud Point	Temp at which wax crystals first appear, making oil cloudy	Important for cold-climate operations; indicates lower temp limit

PROPERTY	DEFINITION	SIGNIFICANCE / UNIT
Pour Point	Lowest temp at which oil just flows under gravity	3°C above the point where oil stops flowing; cold-weather performance
Oiliness	Ability to form a thin adsorbed film on metal surface under boundary lubrication	Important for boundary lubrication
Saponification Value	mg of KOH required to saponify 1g of oil	Indicates amount of fatty acid ester; High SV = more animal/vegetable origin
Acid Value (Neutralisation No.)	mg of KOH needed to neutralize free acids in 1g of oil	Indicates degree of oxidation/rancidity; should be low
Aniline Point	Lowest temp at which equal parts of oil & aniline are completely miscible	Indicates aromatic content; low AP = high aromatics

SOLID LUBRICANTS – WHY USED?

- When liquid lubricants fail at very high temperature
- In vacuum (liquid evaporates)
- Where oil contamination is harmful
- In food machinery & medical equipment

Graphite: Works due to layered structure; layers slide over each other

MoS₂: Works in vacuum (no oxidation needed); better than graphite in vacuum

ADDITIVES IN LUBRICANTS

- **Antioxidants:** Prevent oil degradation
- **Anti-wear agents:** Protect metal surfaces
- **EP Additives:** S, P, Cl compounds
- **VI Improvers:** Polymers that maintain viscosity
- **Pour Point Depressants:** Lower pour point
- **Detergents:** Keep engine clean
- **Corrosion inhibitors:** Protect metal

PORTLAND CEMENT

DEFINITION: Portland cement is a finely ground powder obtained by burning a mixture of limestone (CaCO_3) and clay (containing SiO_2 , Al_2O_3 , Fe_2O_3) at about $1400\text{--}1500^\circ\text{C}$ and grinding the resulting clinker. It sets and hardens by a process of hydration.

Named "Portland" because hardened cement resembles Portland stone (limestone from Dorset, England) — Joseph Aspdin, 1824

CHEMICAL CONSTITUENTS OF PORTLAND CEMENT

CONSTITUENT	CHEMICAL	NOTATION (BOGUE)	%	ROLE / FUNCTION
Tricalcium Silicate	$3\text{CaO}\cdot\text{SiO}_2$	C_3S (Alite)	25–50%	Early strength development; responsible for initial setting & early hardening
Dicalcium Silicate	$2\text{CaO}\cdot\text{SiO}_2$	C_2S (Belite)	25–40%	Late strength (after 28 days); slow hydration; provides long-term durability
Tricalcium Aluminate	$3\text{CaO}\cdot\text{Al}_2\text{O}_3$	C_3A (Aluminate)	5–11%	Flash set (rapid early reaction); attacked by sulfates; causes heat of hydration
Tetracalcium Aluminoferrite	$4\text{CaO}\cdot\text{Al}_2\text{O}_3\cdot\text{Fe}_2\text{O}_3$	C_4AF (Ferrite)	6–10%	Gives cement its grey color; moderate hydration; resists sulfate attack
Calcium Sulfate	$\text{CaSO}_4\cdot 2\text{H}_2\text{O}$ (Gypsum)	CSH_2	2–3%	Controls setting time; prevents flash set of C_3A ; regulates setting rate

RAW MATERIAL OXIDES – COMPOSITION

CaO (Lime) – 60–65%

From limestone (CaCO_3). Provides calcium for silicates. Excess $\text{CaO} \rightarrow$ unsoundness

SiO_2 (Silica) – 17–25%

From clay/sand. Forms C_3S and C_2S . Provides strength

Al_2O_3 (Alumina) – 3–8%

From clay. Forms C_3A . Provides quick setting. Lowers fusion temp

Fe_2O_3 – 0.5–6%

From clay/iron ore. Forms C_4AF . Gives grey colour. Flux in burning

MgO (Magnesia) – <5%

From limestone. Excess $\text{MgO} \rightarrow$ expansion, cracking (unsoundness)

SO_3 – <2%

From gypsum. Controls setting time. Excess $\rightarrow$ expansion

SETTING OF CEMENT

Initial Setting Time: ≥ 30 min (IS code)

Time from adding water to when paste loses plasticity; measured by Vicat needle test

Final Setting Time: ≤ 600 min (10 hours)

Time from adding water to when paste becomes completely hard/rigid

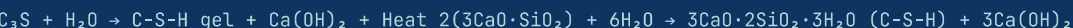
Gypsum role: Reacts with C_3A to form ettringite ($3\text{CaO}\cdot\text{Al}_2\text{O}_3\cdot 3\text{CaSO}_4\cdot 32\text{H}_2\text{O}$) $\rightarrow$ delays rapid setting $\rightarrow$ prevents flash set

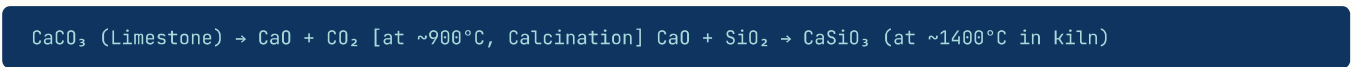
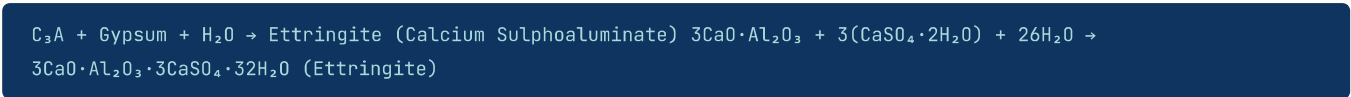
HARDENING OF CEMENT

Hardening = Gain of strength after setting

- $\text{C}_3\text{S} \rightarrow$ fast initial hardening (1–7 days)
- $\text{C}_2\text{S} \rightarrow$ slow, long-term strength (28+ days)
- Hardening due to **hydration** forming **C-S-H gel** (calcium silicate hydrate)
- C-S-H gel fills voids $\rightarrow$ dense paste
- Exothermic process $\rightarrow$ heat of hydration

KEY HYDRATION REACTIONS





STRENGTH DEVELOPMENT TIMELINE

AGE	SOURCE OF STRENGTH	% OF 28-DAY STRENGTH
1 day	Mainly C ₃ A and C ₃ S	~20%
3 days	C ₃ S dominates	~40%
7 days	C ₃ S + C ₂ S starting	~65%
28 days	C ₃ S + C ₂ S	100% (design strength)
1 year+	C ₂ S long-term contribution	125%+

FLASHCARDS – 30 QUICK Q&A

Q: What is a composite material?

Material made of Matrix + Reinforcement; properties better than individual components

Q: Role of reinforcement in composite?

Provides strength, stiffness, and load-bearing capacity to the composite

Q: What is GFRP?

Glass Fibre Reinforced Polymer; Matrix: Polyester; Used: boats, pipes

Q: Classify refractories

Acidic (SiO_2), Basic (MgO), Neutral (Al_2O_3) — based on chemical nature

Q: Highest melting point refractory?

Magnesite (MgO) — $\sim 2800^\circ\text{C}$; Basic refractory

Q: Define lubricant

Substance between moving surfaces to reduce friction, wear, and remove heat

Q: What is Viscosity Index (VI)?

Rate of change of viscosity with temperature; Higher VI = more stable lubricant

Q: Cloud point vs Pour point

Cloud point: wax crystals appear (cloudiness); Pour point: oil stops flowing; Pour point < Cloud point

Q: Boundary lubrication?

Very thin (monomolecular) adsorbed film; occurs at low speed, high load

Q: Why use MoS_2 over graphite in vacuum?

Graphite needs adsorbed gases/moisture to lubricate; MoS_2 works in vacuum without needing gases

Q: Portland cement definition?

Finely ground clinker from burning limestone + clay at $1400\text{--}1500^\circ\text{C}$; hardens by hydration

Q: What is C_2S in cement?

Q: Role of matrix in a composite?

Binds reinforcement, transfers load, protects reinforcement, gives shape

Q: What is a cermet?

Ceramic–Metal composite (e.g., WC in Co matrix); used as cutting tools

Q: Define Refractoriness

Ability to withstand high temperatures (above 1500°C) without melting/softening

Q: Acidic refractory attacked by?

Basic slags (NOT acidic slags — it resists those)

Q: What is spalling in refractories?

Surface peeling/fracture due to thermal gradients during rapid temp changes

Q: What is viscosity?

Resistance of a fluid to flow (internal friction); unit = cP or cSt

Q: Flash point vs Fire point

Flash point: vapours ignite momentarily; Fire point: vapours burn continuously ≥ 5 sec; Fire point > Flash point

Q: Hydrodynamic lubrication?

Thick oil film completely separates surfaces; no metal-to-metal contact; best lubrication

Q: Extreme Pressure (EP) lubrication?

EP additives (S, P, Cl) react with metal to form solid protective film under extreme conditions

Q: What is saponification value?

mg of KOH to saponify 1g of oil; high value = more fatty acid esters = animal/vegetable oil

Q: What is C_3S in cement?

Tricalcium Silicate ($3\text{CaO}\cdot\text{SiO}_2$); gives early strength; responsible for initial setting

Q: Role of gypsum in cement?

Dicalcium Silicate ($2\text{CaO}\cdot\text{SiO}_2$); gives long-term (late) strength; slow hydration

Retards setting time; prevents flash set of C_3A ; forms ettringite with C_3A

Q: What is C-S-H gel?

Calcium Silicate Hydrate; main hydration product; responsible for strength & hardening of cement

Q: Initial vs Final setting time (IS code)?

Initial: ≥ 30 min; Final: ≤ 600 min (10 hours)

Q: What is flash set in cement?

Rapid uncontrollable stiffening of cement due to quick reaction of C_3A ; prevented by gypsum

Q: Ettringite formula?

$3\text{CaO}\cdot\text{Al}_2\text{O}_3\cdot 3\text{CaSO}_4\cdot 32\text{H}_2\text{O}$; formed from C_3A + gypsum + water; controls setting

Q: What gives cement its grey colour?

C_4AF (Tetracalcium Aluminoferrite) — iron oxide compound

Q: Causes of unsoundness in cement?

Excess free CaO (lime), excess MgO , or excess $\text{SO}_3 \rightarrow$ expansion & cracking after hardening

 EXAM TIPS & COMMON MISTAKES **COMPOSITES — WATCH OUT** **MISTAKE 1: CONFUSING MATRIX & REINFORCEMENT**

Matrix = continuous phase (binder). Reinforcement = dispersed phase (strength). Matrix DOES NOT provide primary strength.

 MISTAKE 2: MIXING COMPOSITE TYPES

Particle-reinforced $\neq$ Fibre-reinforced. Key: particles are equiaxed; fibres are elongated (aspect ratio >1). Structural composites = laminates + sandwiches (not about reinforcement type).

 MISTAKE 3: APPLICATION CONFUSION

GFRP (glass) = economical applications (boats, pipes); CFRP (carbon) = high-performance (aerospace, F1). Don't swap these in answers!

 REMEMBER

CMC = best for high temperature + hardness (ceramics);
MMC = high temp + toughness (metals);
PMC = most common, easiest to make (polymers)

 REFRACTORIES — WATCH OUT **MISTAKE 4: ACIDIC/BASIC CONFUSION**

Acidic refractories RESIST acidic conditions but are ATTACKED by basic slags. Always think opposite: name refers to their own chemistry, not what attacks them.

 MISTAKE 5: POROSITY DIRECTION

Higher porosity = WORSE performance (more slag penetration, weaker). Always state: low porosity is desirable in refractories.

 MELTING POINTS TO REMEMBER

MgO (Magnesite): $\sim 2800^\circ\text{C}$ $\rightarrow$ Highest

SiO₂ (Silica): $\sim 1710^\circ\text{C}$

Al₂O₃ (Alumina): $\sim 2050^\circ\text{C}$

Fire clay: $\sim 1840^\circ\text{C}$

 LUBRICANTS — WATCH OUT **MISTAKE 6: FLASH POINT VS FIRE POINT**

Flash point $\rightarrow$ MOMENTARY ignition of vapours (flash). Fire point $\rightarrow$ CONTINUOUS burning ≥ 5 sec. Fire point is ALWAYS higher than flash point. Never reverse!

 MISTAKE 7: CLOUD POINT VS POUR POINT

Cloud point $\rightarrow$ wax crystals appear (OIL BECOMES CLOUDY, still flows). Pour point $\rightarrow$ oil STOPS FLOWING (lower temperature than cloud). Pour point $<$ Cloud point.

 MISTAKE 8: VISCOSITY INDEX

HIGH VI = GOOD (oil viscosity doesn't change much with temp). Students often think high VI = bad. Remember: good quality oils have high VI (>100).

 MISTAKE 9: LUBRICATION MECHANISMS

Hydrodynamic = HIGH speed, LOW load (full film). Boundary = LOW speed, HIGH load (thin film). Extreme Pressure = HIGHEST load (chemical reaction film). Sequence: HD $\rightarrow$ Mixed $\rightarrow$ Boundary $\rightarrow$ EP.

 CEMENT — WATCH OUT **MISTAKE 10: C₃S VS C₂S ROLE**

C₃S $\rightarrow$ EARLY strength (1–7 days). C₂S $\rightarrow$ LATE/long-term strength (28+ days). Never mix these — C₂S does NOT give early strength!

 MISTAKE 11: SETTING VS HARDENING

Setting = loss of plasticity (cement paste stiffens). Hardening = gain of strength (after setting). These are DIFFERENT processes — don't use interchangeably!

 MISTAKE 12: GYPSUM ROLE

Gypsum RETARDS (slows) setting — it does NOT accelerate it. It forms ettringite by reacting with C₃A $\rightarrow$ prevents flash set. Without gypsum $\rightarrow$ instant unusable hardening.

 BOGUE NOTATION MEMORY TRICK

C = CaO, S = SiO₂, A = Al₂O₃, F = Fe₂O₃, H = H₂O

C₃S, C₂S, C₃A, C₄AF — these four clinker compounds are the CORE of cement chemistry.

⚡ QUICK REVISION SHEET — ALL DEFINITIONS & KEY FACTS

⚙️ COMPOSITES

Composite	Matrix + Reinforcement = enhanced properties
Matrix	Continuous binder phase; transfers load
Reinforcement	Dispersed phase; provides strength/stiffness
PMC	Polymer Matrix Composite; most common; GFRP, CFRP
MMC	Metal Matrix Composite; high-temp; Al+SiC
CMC	Ceramic Matrix Composite; hardest; turbine blades
GFRP	Glass + Polyester; boats, pipes; cheap
CFRP	Carbon + Epoxy; aerospace; expensive, strongest
Cermet	WC + Co; ceramic + metal; cutting tools
Laminate	Layers bonded together; plywood
Sandwich	Stiff faces + light core; honeycomb panels

🔥 REFRACTORIES

Refractory	Withstands T > 1500°C; inert
Acidic	SiO ₂ -based; resists acid slags; Silica brick
Basic	MgO-based; resists basic slags; Magnesite
Neutral	Al ₂ O ₃ /Cr ₂ O ₃ ; resists both; Fire clay
Refractoriness	Ability to resist fusion at high temp
Spalling	Surface peeling due to thermal shock
Low porosity	Desired → less slag penetration
PCE	Pyrometric Cone Equivalent; measures refractoriness
MgO m.p.	~2800°C; highest melting basic refractory
Fire clay	35–45% Al ₂ O ₃ ; neutral; cheap; most used

📊 LUBRICANTS

Lubricant	Reduces friction, wear; removes heat
Viscosity	Resistance to flow; cP or cSt; critical property
VI	Viscosity Index; higher = better stability with temp
Flash Pt	Vapours ignite momentarily; safety limit
Fire Pt	Burns ≥5 sec; > Flash point always
Cloud Pt	Wax crystals appear; upper cold-flow limit
Pour Pt	Oil stops flowing; 3°C above no-flow point
Sapon. V.	mg KOH / g oil; high = animal/veg oil
Acid Value	Free acidity in oil; should be low
Hydrodynamic	Full film; high speed + low load; best
Boundary	Mono-layer film; low speed + high load
EP Lubr.	Chemical film; extreme load; S/P/Cl additives
Graphite	Solid lubricant; layered; works in air
MoS₂	Solid lubricant; layered; works in vacuum

🏠 PORTLAND CEMENT

Cement	Clinker ground to powder; hardens by hydration
C₃S (Alite)	25–50%; early strength; fast hydration
C₂S (Belite)	25–40%; late strength; slow hydration
C₃A	5–11%; flash set; most heat; sulfate sensitive
C₄AF	6–10%; grey colour; moderate hydration
Gypsum	2–3%; retarder; prevents flash set
C-S-H gel	Main product of hydration; responsible for strength
Ettringite	C ₃ A + Gypsum + H ₂ O; controls initial set
Initial set	≥30 min (IS code); Vicat test
Final set	≤600 min; completely rigid
Hydration	Cement + water → heat + C-S-H + Ca(OH) ₂
Unsoundness	Excess CaO / MgO / SO ₃ → expansion, cracks
Calcination	CaCO ₃ → CaO + CO ₂ at ~900°C
28-day str.	Design strength; both C ₃ S + C ₂ S contribute

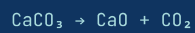
📐 ALL KEY FORMULAS & REACTIONS AT A GLANCE

Hydration (C₃S):
 $2\text{C}_3\text{S} + 6\text{H}_2\text{O} \rightarrow$
 $\text{C-S-H} + 3\text{Ca(OH)}_2$

Hydration (C₂S):
 $2\text{C}_2\text{S} + 4\text{H}_2\text{O} \rightarrow$
 $\text{C-S-H} + \text{Ca(OH)}_2$

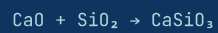
Gypsum reaction:
 $\text{C}_3\text{A} + 3(\text{CaSO}_4 \cdot 2\text{H}_2\text{O}) + 26\text{H}_2\text{O}$
 $\rightarrow \text{Ettringite}$

Calcination:



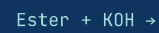
(~900°C)

Clinker burning:



(~1400–1500°C)

Saponification:



Soap + Alcohol



LAST-MINUTE MEMORY HOOKS:

$\text{C}_3\text{S} \rightarrow$ Early $\text{C}_2\text{S} \rightarrow$ Late $\text{C}_3\text{A} \rightarrow$ Flash set (gypsum tames it) Flash < Fire Pour < Cloud High VI = Good oil Acidic refractory ≠ Acid-resistant MoS_2 = Vacuum lubricant